

RE-AMP: ROBUSTNESS EVALUATION OF ADVERSARIAL MUSIC POISONING

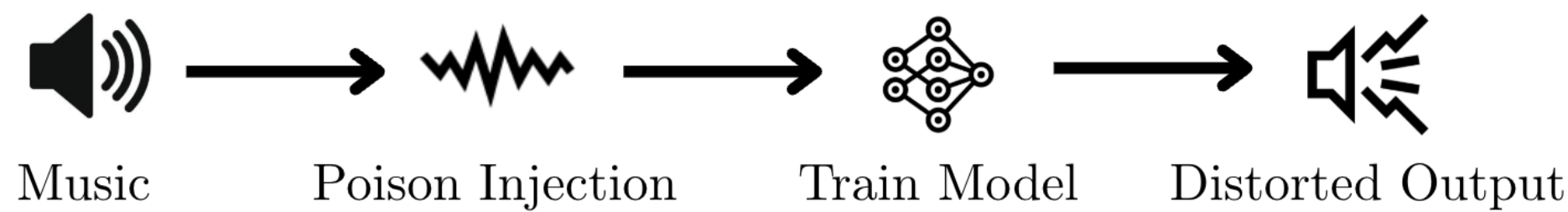
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Motivation & Overview



Generative music models based on neural codecs and audio language models are rapidly improving [1, 2], yet there is still no standard way to measure their vulnerability to data poisoning, despite extensive work on clean-label attacks in vision [5]. Existing audio attacks rarely account for codec tokenization, long-range musical structure, or common production transforms (pitch shifts, time-stretching, lossy compression), making it unclear which poisons survive realistic pipelines and which defenses preserve musical quality. RE-AMP (Robustness Evaluation of Adversarial Music Poisoning) addresses this gap with a music-specific robustness benchmark for clean-label poisoning. Using MusicGen-small on the Song Descriptor Dataset [4], we compare a codec- and music-aware poison, HarmonyDagger, to naïve Gaussian baselines under realistic “removal” pipelines and quantify their impact on optimization dynamics, generative drift, and perceptual audio quality via Fréchet Audio Distance (FAD) [3].

Framework & Threat Model

RE-AMP models an interaction between an adversarial creator and a dataset curator training a neural codec-based music generator. Starting from a clean corpus D_{clean} of waveform tracks, an attack operator P_ϕ adds imperceptible, clean-label perturbations to a subset of creator-owned songs, producing a poisoned set D_{poison} with unchanged metadata and labels. We instantiate P_ϕ either as naïve additive Gaussian noise or as *HarmonyDagger*, a codec- and music-aware poison that operates in the time-frequency domain, aligns perturbations with EnCodec frame boundaries [2], and enforces simple psychoacoustic bounds so that x and $x + \delta$ remain plausible to typical listeners. On the defender side, a removal operator R_j is applied to incoming audio before training; for each condition (clean, poisoned, and poison+removal) we fine-tune MusicGen-small [1], obtaining a model $f_\theta^{(j)}$. This P_ϕ - R_j - f_θ abstraction decouples attack, defense, and architecture and can be reused with other codecs, datasets, and generative audio systems.

Poison Removal Operators

In RE-AMP, we treat poison removal as an approximate inverse to the attack: given audio that may contain Gaussian or HarmonyDagger poisons, a curator applies a transform R_j before training to recover cleaner behavior.

Black-box transforms are common, attack-agnostic signal edits. Pitch shifting (± 2 semitones) tests whether perturbations tied to absolute frequency survive modest key changes, and down-up resampling ($44.1 \rightarrow 22.05 \rightarrow 44.1$ kHz) mimics bandwidth-limited normalization or streaming pipelines.

White-box transforms are structure-aware and tuned to HarmonyDagger’s time-frequency footprint. STFT median filtering suppresses narrow-band, locally inconsistent artifacts while preserving coherent musical energy. Adaptive time-frequency masking assumes access to an approximate poison mask $M(t, f)$ and softly attenuates high-confidence regions in the spectrogram. We evaluate each R_j by fine-tuning MusicGen-small on cleaned datasets and comparing CE, PPL, GradNorm, GD, and FAD [3] against clean and poisoned baselines.

Results

We fine-tune MusicGen-small on clean, Gaussian-poisoned, and HarmonyDagger-poisoned variants of the Song Descriptor Dataset [4], with and without removal transforms. Relative to clean, Gaussian poisons mildly worsen optimization and quality: CE and PPL increase, gradient norms grow, GD becomes non-zero, and FAD rises but remains moderate. HarmonyDagger produces substantially stronger effects: training becomes less stable than in both clean and Gaussian conditions, GradNorm is larger, GD more than doubles relative to Gaussian, and FAD reaches the highest values, indicating heavier optimization drag and larger style shift in generated audio. White-box removals partially undo these effects. STFT median filtering and adaptive time-frequency masking move loss curves closer to the clean trajectory and reduce GD and FAD for HarmonyDagger, with adaptive masking giving the strongest recovery but not fully closing the gap. In contrast, black-box signal edits such as pitch shifting and down-up resampling provide weaker mitigation: they modestly reduce perceptual artifacts but leave training dynamics and GD closer to the poisoned condition, suggesting that common production and streaming pipelines alone are insufficient to neutralize codec- and music-aware poisons.



Fig. 2: Training loss under poisoning and removal.

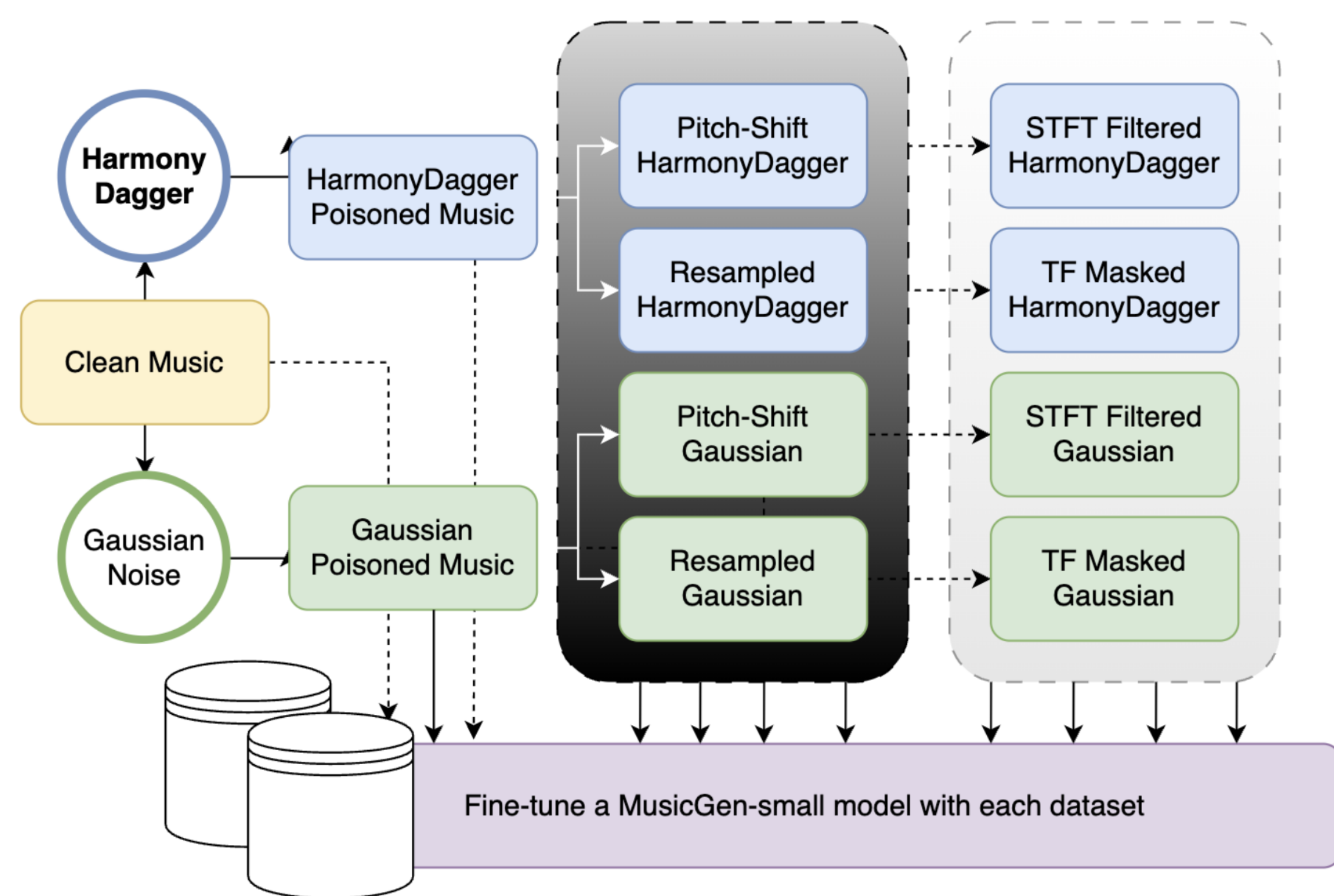


Fig. 3: RE-AMP fine-tuning pipeline.

Quantitative Comparison

Condition	CE ↓	PPL ↓	GradNorm ↓	GD ↑	FAD ↓
Clean	3.48	32.5	2.41	0.00	1.12
Gaussian (baseline)	3.61	38.9	2.97	1.82	1.94
HarmonyDagger	3.82	45.6	3.67	4.53	2.87

Removal	Gaussian poison				HarmonyDagger poison			
	CE	PPL	GD	FAD	CE	PPL	GD	FAD
STFT median	3.64	40.2	2.11	1.81	3.67	41.1	2.91	1.96
Adaptive TF mask	3.58	38.9	1.88	1.72	3.59	38.8	2.44	1.84
Pitch shift	3.71	43.0	2.36	2.04	3.74	43.7	3.18	2.21
Down-up resamp.	3.66	41.9	2.22	1.95	3.69	42.5	3.02	2.15

HarmonyDagger sharply increases CE, PPL, GradNorm, GD, and FAD over both clean and Gaussian baselines, indicating stronger optimization drag and style shift. White-box removals (STFT median, adaptive TF masking) move HarmonyDagger closest to clean, especially in GD and FAD, while black-box edits (pitch shift, down-up resampling) yield smaller, less consistent gains and never fully close the gap. Each entry reflects a single training run and is intended to illustrate trends rather than provide precise estimates with confidence intervals.

Remarks

RE-AMP is currently limited to a single model family (MusicGen-small), dataset regime, and embedding backbone, and each condition is trained only once, so reported metrics should be read as illustrative, not definitive. Still, consistent gaps between clean, Gaussian, and HarmonyDagger conditions, and between black-box and white-box removals, suggest that codec-aware poisons and structure-aware defenses deserve closer study across architectures, genres, and human listening tests.

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References

- [1] Jade Copet et al. “Simple and Controllable Music Generation”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. URL: https://proceedings.neurips.cc/paper_files/paper/2023/file/94b472a1842cd7c56dcb125fb2765fbd-Paper-Conference.pdf.
- [2] Alexandre Défossez et al. “High Fidelity Neural Audio Compression”. In: *arXiv:2210.13438* (2022). URL: <https://arxiv.org/abs/2210.13438>.
- [3] Kevin Kilgour et al. *Fréchet Audio Distance: A Metric for Evaluating Music Enhancement Algorithms*. 2019. arXiv: 1812.08466 [eess.AS]. URL: <https://arxiv.org/abs/1812.08466>.
- [4] Ilaria Manco et al. *The Song Descriptor Dataset: a Corpus of Audio Captions for Music-and-Language Evaluation*. 2023. arXiv: 2311.10057 [cs.SD]. URL: <https://arxiv.org/abs/2311.10057>.
- [5] Ali Shafahi et al. “Poison Frogs! Targeted Clean-Label Poisoning Attacks on Neural Networks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2018. URL: https://proceedings.neurips.cc/paper_files/paper/2018/hash/22722a343513ed45f14905eb07621686-Abstract.html.